On the Adams Spectral Sequence and the Hopf invariant one problem.

Thm (Hopf invariant one, Adams '58)

For $n \in \mathbb{Z}_{\geq 0}$, TFAE:

(1) $n = 0, 1, 3, 7$

(2) $n = 2^j - 1$, and in the mod 2 Adams SS of the sphere the element h_j is a permanent cycle.

(3) There exist a stably framed n-manifold M^n w/ self-linking number 1 mod 2.

[Pontryagin-Thom construction $\rightsquigarrow$ $S^{2n+1} \xrightarrow{f} S^{n+1}$, $M^n = f^{-1}$(some regular value) $\rightsquigarrow$ self-linking number of M^n in S^{2n+1}]

(4) There exist a map $f: S^{2n+1} \longrightarrow S^{n+1}$ s.t. $H^*(C_f = S^{n+1} \cup_{S^{2n+1}} D^{2n+2}; \mathbb{F}_2) \cong \mathbb{F}_2[x]/x^3$, $|x| = n+1$

(5) S^n is parallelizable. (w/ the standard smooth structure)

(6) S^n admits an A_2-structure $S^n \vee S^n \xrightarrow{\text{fold}} S^n$, $S^n \vee S^n \hookrightarrow S^n \times S^n$, $S^n \times S^n \dashrightarrow S^n$

(7) $\mathbb{R}^{n+1}$ admits an $\mathbb{R}$-division-algebra structure (not necessarily associative)

Adams SS:

$$E_2^{n,s} \cong Ext_{A^*}^{s,s+n}(\mathbb{F}_2, \mathbb{F}_2) \Longrightarrow \pi_n(\mathbb{S}_2^\wedge)$$

- A^* : mod 2 Steenrod Alg.
- $Ext_{A^*}^{s,s+n}(-,-)$: the Ext group in graded A^*-mod.

  s : homological degree / filtration degree

  $s+n$: internal degree

  n : topological degree / stem

- $\pi_n(\mathbb{S}_2^\wedge) \cong \pi_n(\mathbb{S})_2^\wedge \cong \begin{cases} \mathbb{Z}_2, & n=0 \\ 2\text{-primary part of } \pi_n(\mathbb{S}) & n\neq 0 \end{cases}$

- multiplicative structure:
  - $E_2^{*,*}$ is a graded-commutative $\mathbb{F}_2$-alg.
    [i.e. $x\cdot y = y\cdot x$]
  - $\forall\, 3\le r\le\infty$, E_r^{**} is also a graded-commutative $\mathbb{F}_2$-alg as a subquotient of E_2^{**}.
    [i.e, $E_r^{**} \cong Z_r^{**}/B_r^{**}$ $\quad Z_r^{**}\le E_2^{**}$ is a sub-alg. $\quad B_r^{**}\le Z_r^{**}$ is an ideal.]
  - $\forall\, 2\le r<\infty$, $d_r(xy) = d_r(x)\cdot y + x\cdot d_r(y)$
  - As the 2-completion of $\pi_*(\mathbb{S})$, $\pi_*(\mathbb{S}_2^\wedge)$ is a graded-commutative ring, [$x\cdot y = (-1)^{|x|\cdot|y|} y\cdot x$]

$$E_\infty^{*,*} \cong gr_* \pi_*(\$)$$

is an isom. of graded-commutative rings.

Some nontrivial facts on $E_2^{n,s} \cong Ext_{A^*}^{s,s+n}(\mathbb{F}_2, \mathbb{F}_2)$

- $n<0$, $E_2^{n,s} \cong 0$
- $n=0$, $E_2^{0,*} \cong \mathbb{F}_2[h_0]$, $h_0 \in E_2^{0,1}$
- $n>0$, $E_2^{n,s} \cong 0$ if $s \geq \frac{1}{2}n+2$
- In lower filtration degrees $s \leq 5$, there is an explicit description of $E_2^{*,s}$ in terms of generators & relations
  - $s \leq 2$ Adams '58
  - $s=3$ Wang '67
  - $s=4$ Lin '08
  - $s=5$ Chen '11
  - $s \geq 6$ unknown

Thm There exist non-zero elements

$$h_j \in E_2^{2^j-1,1}, \quad C_j \in E_2^{2^{j+3}+2^{j+1}+2^j-3,3}, \quad j \geq 0$$

s.t.

(*)
$$\begin{cases} h_j \cdot h_{j+1} = 0, & j \geq 0 \\ h_j h_{j+2}^2 = 0, & j \geq 0 \\ h_{j+1}^3 = h_j^2 h_{j+2}, & j \geq 0 \\ h_j^2 \cdot h_{j+3}^2 = 0, & j \geq 0 \\ C_i \cdot h_j = 0, & j = i-1, i, i+2, i+3, \ i,j \geq 0 \end{cases}$$

Moreover, the induced ring homom.

$$\mathbb{F}_2[h_j, c_j : j \geq 0]/(*) \longrightarrow E_2^{*,*}$$

is bijective $s \leq 3$

and injective $s = 4$

$\Rightarrow$ In $E_2^{*,0}$, $\mathbb{F}_2$-basis 1

$E_2^{*,1}$, $\mathbb{F}_2$-basis $\{h_j : j \geq 0\}$

$E_2^{*,2}$, $\mathbb{F}_2$-basis $\{h_j^2 : j \geq 0\} \amalg \{h_i h_j : j \geq i+2\}$

$E_2^{*,3}$, $\mathbb{F}_2$-basis $\{h_j^3 : j \geq 0\} \amalg \{h_i^2 h_j : \begin{smallmatrix} i,j \geq 0 \\ j \notin [i-2, i+2] \end{smallmatrix}\}$

$\{h_i h_j h_k : \begin{smallmatrix} i \geq 0 \\ j \geq i+2 \\ k \geq i+1 \end{smallmatrix}\} \amalg \{c_j : j \geq 0\}$

$E_2^{*,4}$, $\mathbb{F}_2$-linearly independent set.

$\{h_0^4\} \amalg \{h_i^3 h_j : i, j \geq 0,\ j \notin [i-3, i+2]\}$

$\{h_i^2 h_j^2 : i \geq 0,\ j \geq i+4\} \amalg$

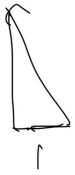
$\{h_i^2 h_j h_k : i \geq 0,\ j \geq i+3,\ k \geq j+1\}$

$\{h_i h_j^2 h_k : i \geq 0,\ j \geq i+3,\ k \geq j+3\}$

$\{h_i h_j h_k^2 : i \geq 0,\ j \geq i+2,\ k \geq j+3\}$

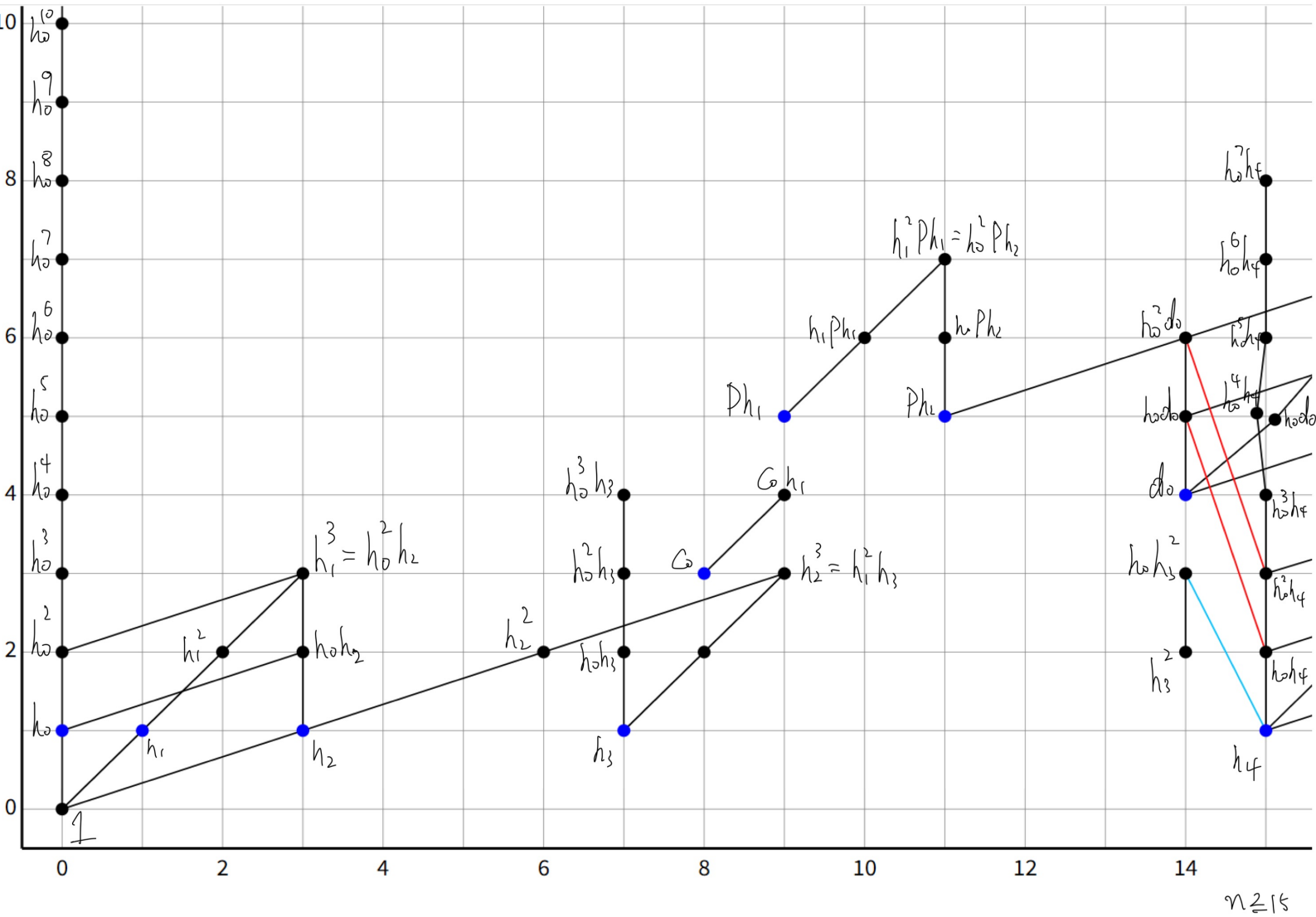
$\{h_i h_j h_k h_\ell : i \geq 0,\ j \geq i+2,\ k \geq j+2,\ \ell \geq k+2\}$

$\{c_i \cdot h_j : i, j \geq 0,\ j \notin \{i-1, i, i+2, i+3\}\}$

$\bullet$: $\mathbb{F}_2$, d_r : r 

: multi. by h_0

: multi. by h_1



Thm h_0, h_1, h_2, h_3 are perpanent cycles.

$$d_2(h_j) = h_0 h_{j-1}^2 \qquad j \geq 4$$

Proof (Wang '67) Induction on j.

h_0, h_2, h_3 permanent cycles for degree reasons.

We prove $d_r(d_1) = 0$ inductively on r

$h_0 h_1 = 0 \Rightarrow 0 = d_r(h_0 h_1) = d_r(h_0) h_1 + h_0 d_r(h_1)$

$\Rightarrow h_0 \cdot d_r(h_1) = d_r(h_0) h_1 = 0 \Rightarrow d_r(h_1) = 0$

$\Rightarrow h_1$ permanent cycle.

$h_0 \in E_\infty^{0,1}$ ⟿ $2 \in \pi_0(\mathbb{S}_2^\wedge) \cong \mathbb{Z}_2^\wedge$

$h_1 \in E_\infty^{1,1}$ ⟿ $\eta \in \pi_1(\mathbb{S}_2^\wedge)$

$h_2 \in E_\infty^{3,1}$ ⟿ $\nu \in \pi_3(\mathbb{S}_2^\wedge)$

$h_3 \in E_\infty^{7,1}$ ⟿ $\sigma \in \pi_7(\mathbb{S}_2^\wedge)$

- $j = 4$, $|\sigma| = 7 \Rightarrow \sigma \cdot \sigma = -\sigma \cdot \sigma$

$\Rightarrow 2 \cdot \sigma \cdot \sigma = 0 \in \pi_{14}(\mathbb{S}_2^\wedge)$

$\Rightarrow h_0 \cdot h_3^2 = 0$ in $E_\infty^{14,3}$

$\Rightarrow d_2(h_4) = h_0 h_3^2$

$j \geq 5$, $d_2(h_{j-1} \cdot h_j) = 0$

$\| $

$d_2(h_{j-1}) \cdot h_j + h_{j-1} d_2(h_j)$

$\Rightarrow h_{j-1} d_2(h_j) = d_2(h_{j-1}) \cdot h_j$ ↙ induction on j

$= h_0 h_{j-2}^2 h_j$

$= h_0 h_{j-1}^3 \neq 0$

$\Rightarrow d_2(h_j) \neq 0 \in E_2^{2^j-2,3} \cong \mathbb{F}_2\{h_0 h_{j-1}^2\}$

$\Rightarrow d_2(h_j) = h_0 h_{j-1}^2$

17